THE HIPSTER CISO

hc-cdaio-kg: Knowledge Graph Architecture and Schema Reference

hc-cdaio-kg: Knowledge Graph Architecture and Schema Reference: Technical Document

Thomas Jones

Comprehensive technical reference documenting the data architecture, entity type schemas, relationship patterns, and structural design of the hc-cdaio-kg enterprise knowledge graph. Covers 2,944 entities across 26 types, 7,466 relationships across 42 types, and 401 structured due diligence analysis files.

Publication Date: 2026-03-04

https://www.thehipsterciso.com

Contents

1 Platform and Dataset 1

1.1 hc-enterprise-kg: The Platform 1

1.2 hc-cdaio-kg: The Dataset 1

2 Architecture Overview 3

2.1 Design Philosophy 3

2.2 Graph Statistics 3

3 Base Entity Schema 6

3.1 Identity and Classification 7

3.2 Temporal Metadata 7

3.3 Provenance 7

3.4 Temporal Object 7

4 Entity Type Catalog 8

4.1 Operational Infrastructure 8

`role` (427 entities) 8 • `integration` (378 entities) 10 • `control` (340 entities) 12

4.2 Organizational Structure 13

`department` (250 entities) 13 • `organizational_unit` (109 entities) 14 • `person` (100 entities) 17

4.3 Data and Information 19

`data_asset` (164 entities) 19 • `data_domain` (88 entities) 23

4.4 Systems and Technology 25

`system` (164 entities) 25 • `network` (94 entities) 29 • `site` (58 entities) 30 • `location` (46 entities) 32

4.5 Market and External 32

`customer` (85 entities) 33 • `vendor` (55 entities) 34 • `regulation` (70 entities) 35 • `jurisdiction` (61 entities) 36 • `geography` (57 entities) 37 • `contract` (37 entities) 38 • `market_segment` (26 entities) 39

4.6 Strategic and Risk 39

`initiative` (24 entities) 40 • `risk` (22 entities) 41 • `business_capability` (64 entities) 44 • `incident` (7 entities) 45

4.7 Products and Services 46

`product_portfolio` (71 entities) 46 • `product` (33 entities) 47

4.8 Governance 48

`policy` (114 entities) 48

5 Relationship Architecture 50

5.1 Base Relationship Schema 50

5.2 Spatial and Jurisdictional 50

`located_in` (1260 edges) 51 • `located_at` (683 edges) 51

5.3 Dependency and Constraint 52

`subject_to` (1120 edges) 52 • `depends_on` (790 edges) 52 • `hosted_on` (490 edges) 53 • `applies_to` (543 edges) 53

5.4 Governance and Control 53

`enforces` (386 edges) 53 • `managed_by` (385 edges) 54 • `implements` (178 edges) 54 • `governs` (32 edges) 54 • `governed_by` (12 edges) 55

5.5 Data Flow 55

`stores` (165 edges) 55 • `connects_to` (140 edges) 55 • `integrates_with` (101 edges) 55 • `processes` (55 edges) 56

5.6 Organizational 56

`has_role` (92 edges) 56 • `works_in` (48 edges) 56 • `reports_to` (9 edges) 56 • `staffed_by` (3 edges) 57 • `belongs_to` (71 edges) 57 • `holds` (37 edges) 57 • `contains` (75 edges) 57

5.7 Commercial 58

`serves` (137 edges) 58 • `provides_service` (71 edges) 58 • `buys` (22 edges) 58 • `contracts_with` (7 edges) 59 • `supplies` (6 edges) 59 • `supplied_by` (28 edges) 59 • `delivers` (33 edges) 59

5.8 Risk and Impact 60

`mitigates` (29 edges) 60 • `creates_risk` (4 edges) 60 • `impacts` (14 edges) 60 • `impacted_by` (17 edges) 60 • `affects` (21 edges) 61 • `addresses` (15 edges) 61

5.9 Strategic 61

`supports` (209 edges) 61 • `enables` (33 edges) 62 • `drives` (28 edges) 62 • `funded_by` (12 edges) 62 • `subject_of` (30 edges) 63 • `hosts` (26 edges) 63 • `regulates` (49 edges) 63

6 Edge Pattern Analysis 64

6.1 Relationship Volume Distribution 64

6.2 Entity Type Connectivity 64

7 Due Diligence Analysis Layer 67

7.1 Analysis File Structure 68

8 Known Structural Gaps 69

8.1 Data Quality Targets 69

8.2 Governance Relationship Density 69

8.3 Temporal Window Coverage 69

1 Platform and Dataset

1.1 hc-enterprise-kg: The Platform

The [hc-enterprise-kg](https://github.com/thehipsterciso/hc-enterprise-kg) is an open-source enterprise knowledge graph platform designed to generate and analyze structural models of organizations. It provides the schema definition (30 entity types, 58 relationship types), the Pydantic v2 data models, the NetworkX graph backend, and a 16-tool MCP server that exposes graph analysis capabilities to AI assistants through natural language.

The platform's core premise is that in enterprise decision-making, the relationships between entities are where the insight lives. A list of systems is inventory. A graph of systems connected by dependency, hosting, governance, and data flow relationships is intelligence. hc-enterprise-kg provides the schema vocabulary and analytical tooling to build and interrogate that graph.

The platform supports synthetic generation for scenario modeling (with industry-specific profiles for technology, financial services, and healthcare organizations), CSV and text ingestion for real-world data, and a quality validation module that scores schema compliance, relationship integrity, and field correlation consistency. The MCP server enables Claude Desktop to perform blast radius analysis, centrality scoring, shortest-path traversal, and entity search against any loaded graph through conversational interaction.

1.2 hc-cdaio-kg: The Dataset

This document describes hc-cdaio-kg, which is a specific dataset built on the hc-enterprise-kg platform. Where hc-enterprise-kg defines how to model an enterprise, hc-cdaio-kg models a specific one: Rackspace Technology, Inc., constructed entirely from public sources using AI-augmented OSINT as a practicum deliverable for the Carnegie Mellon University Heinz College CDAIO program.

The relationship between platform and dataset is analogous to a database engine and a specific database instance. hc-enterprise-kg provides the schema, validation rules, and query tools. hc-cdaio-kg provides the populated entities, relationships, and analytical conclusions for one target organization. The platform's 30-type schema defines the maximum vocabulary; this dataset uses 26 of those types and 42 of the 58 available relationship types, reflecting what was discoverable and material for the target organization.

The dataset includes a structured due diligence analysis layer (401 JSON files across nine stages) that does not exist in the platform itself. This layer represents the analytical methodology and findings specific to evaluating Rackspace Technology as a potential acquisition, investment, or transformation target. The platform provides the structural graph; hc-cdaio-kg adds both the populated data and the assessed conclusions.

Concern	hc-enterprise-kg (Platform)	hc-cdaio-kg (Dataset)
Schema definition	30 entity types, 58 relationship types	26 entity types, 42 relationship types (subset)
Data population	Synthetic generation or ingestion	OSINT-sourced from public filings and records
Graph backend	NetworkX (swappable to Neo4j)	NetworkX via assembled graph.json
MCP server	16 analytical tools	Loaded as the active graph instance
Quality validation	Schema compliance, relationship integrity	Pre-commit hooks, GitHub Actions CI
Analytical layer	Not included	401 JSON files across 9 due diligence stages
Target audience	Any enterprise modeling use case	Rackspace Technology CDAIO assessment

2 Architecture Overview

The hc-cdaio-kg knowledge graph is an enterprise intelligence structure built to model a single target organization across every dimension material to a CDAIO-grade due diligence assessment. The graph stores 2,944 entities connected by 7,466 relationships, encoding 26 entity types and 42 relationship types that form 114 distinct source-type / relationship-type / target-type edge patterns.

The graph is accompanied by 401 structured JSON analysis files organized across nine due diligence stages, which represent the analytical conclusions built on top of the structural data.

2.1 Design Philosophy

Three principles govern the schema design:

Attribute density over entity count. The schema prioritizes deep attribute coverage within each entity type over creating large populations of shallow entities. Data assets carry 60+ fields at near-complete population. In a due diligence context where follow-up access to the target organization may be limited, that density determines whether the graph generates new questions or answers existing ones.

Temporal versioning. Every entity carries `created_at`, `updated_at`, `valid_from`, `valid_until`, and `version` fields as part of the base schema. The graph can represent enterprise state at a specific point in time and track how entities change, which matters when assessing whether a governance program is improving or stagnating.

Provenance tracking. Every entity includes a `provenance` field recording the source and quality assessment of the information used to create it. In due diligence, the difference between a fact, an inference, and a claim from management can determine the reliability of the entire assessment. Entity-level provenance makes the graph's own epistemic confidence queryable.

2.2 Graph Statistics

Entity type distribution ordered by population count:

Entity Type	Count	% of Total	Schema Fields
role	427	14.5%	63
integration	378	12.8%	32

Entity Type	Count	% of Total	Schema Fields
control	340	11.5%	35
department	250	8.5%	11
data_asset	164	5.6%	83
system	164	5.6%	117
policy	114	3.9%	36
organizational_unit	109	3.7%	86
person	100	3.4%	70
network	94	3.2%	12
data_domain	88	3.0%	23
customer	85	2.9%	60
product_portfolio	71	2.4%	20
regulation	70	2.4%	21
business_capability	64	2.2%	81
jurisdiction	61	2.1%	23
site	58	2.0%	83
geography	57	1.9%	20
vendor	55	1.9%	81
location	46	1.6%	15
contract	37	1.3%	43
product	33	1.1%	72
market_segment	26	0.9%	24
initiative	24	0.8%	79
risk	22	0.7%	43

Entity Type	Count	% of Total	Schema Fields
incident	7	0.2%	19

3 Base Entity Schema

Every entity in the graph inherits a common set of base attributes regardless of type. These attributes provide identity, temporal context, versioning, and provenance tracking across the entire graph.

Attribute	Population %	Sample Value
id	100.0%	bc-001
entity_type	100.0%	business_capability
created_at	100.0%	2026-02-25T19:01:57.283991Z
updated_at	100.0%	2026-03-04T08:43:28.787752Z
valid_from	0.0%	None
valid_until	0.0%	None
version	100.0%	2
tags	0.0%	[]
name	100.0%	Corporate Strategy & Planning
description	97.5%	Enterprise-level strategic planning, long-term vision develo
provenance	100.0%	{'data_quality_score': {'completeness_pct': 50.0, 'accuracy_
temporal	100.0%	{'effective_date': None, 'expiration_date': None, 'last_revi

3.1 Identity and Classification

The `id` field is a unique identifier (format varies by type: some use structured IDs like `bc-001` for business capabilities, others use UUIDs). The `entity_type` field classifies each entity into one of 26 types, which determines which type-specific attributes are available. The `name` field provides a human-readable label, populated across 100% of entities. The `description` field carries a prose summary of the entity's purpose, populated at 97.5%.

3.2 Temporal Metadata

The `created_at` and `updated_at` fields track when each entity was first ingested into the graph and when it was last modified (both ISO 8601 timestamps, 100% populated). The `valid_from` and `valid_until` fields are available for modeling the business-effective period of an entity but are currently unpopulated (0%), indicating the graph captures current-state rather than historical-state data. The `version` field (100% populated) tracks modification count, enabling change detection across graph iterations.

3.3 Provenance

The `provenance` field is a structured object containing data quality scores (completeness percentage, accuracy confidence, timeliness score) and source attribution. This field is populated at 100% across all 2,944 entities. Provenance operates at the entity level rather than the attribute level, meaning the quality assessment applies to the entity as a whole rather than to individual fields.

3.4 Temporal Object

The `temporal` field is a structured object carrying `effective_date`, `expiration_date`, `last_review_date`, and `next_review_date`. While the field itself is populated at 100%, the interior dates may be null for entities where review cadence has not been established. This structure exists to support governance maturity tracking: an entity with a populated `next_review_date` is under active stewardship, while one with null review dates may represent an unmanaged asset.

4 Entity Type Catalog

This chapter documents every entity type in the graph, organized by functional layer. Each type section includes a rationale for the layer's existence, a complete attribute schema table showing field names and population percentages, and sample values where they illuminate the schema's intent.

4.1 Operational Infrastructure

These entity types model the connective tissue of enterprise operations. Roles define who has authority over what. Integrations define how systems exchange data. Controls define what governance mechanisms exist and whether they function. Most due diligence assessments treat these as second-order concerns behind financials and market position. This graph treats them as primary because operational fragility is where post-acquisition surprises live.

4.1.1 `role` (427 entities)

Field	Populated	Sample Value
access_level	100.0%	privileged
authority_delegated	100.0%	{'financial_approval_limit': {'amount': None, 'currency
competency_model_reference	100.0%	{'model_name': '', 'applicable_competencies': []}
department	0.2%	dept-218
description	99.8%	Top executive responsible for overall corporate strateg
direct_reports_target	100.0%	{'target_count': None, 'actual_count': None, 'includes_
fte_count	0.2%	1
is_active	0.2%	True

Field	Populated	Sample Value
is_privileged	100.0%	True
name	100.0%	Chief Executive Officer
physical_requirements	100.0%	{'has_physical_requirements': False, 'requirement_descr
provenance	100.0%	{'data_quality_score': {'completeness_pct': 45.0, 'accu
reports_to	0.2%	role-001
required_education	100.0%	{'minimum_level': '', 'preferred_fields': [], 'equivale
required_experience	100.0%	{'minimum_years_total': None, 'minimum_years_in_domain'
responsibilities	0.2%	Oversees product strategy across Private and Public Clo
role_family	100.0%	{'family_name': '', 'subfamily': ''}
role_id	99.8%	ROLE-001
role_level	100.0%	{'level_code': 'M1', 'level_name': 'Management', 'level
role_mandate_document	100.0%	{'document_reference': '', 'document_date': None, 'last
role_type	100.0%	Executive
seniority	0.2%	C-Level
temporal	100.0%	{'effective_date': None, 'expiration_date': None, 'last
travel_requirement	100.0%	{'travel_pct': None, 'travel_scope': '', 'travel_freque

Unpopulated fields (0%): allocation_model, belongs_to_unit, compensation_range, decision_rights, department_id, dotted_line_to, employment_type_target, filled_by_persons, fills_capability, functional_domain_primary, functional_domain_secondary, governance_memberships, governed_by, headcount_authorized, headcount_by_location, headcount_filled, headcount_vacant, language_requirements, location_flexibility, max_headcount, metadata, origin, permissions, regulatory_accountability, regulatory_designation, reports_to_role, required_certifications, required_clearances, required_skills, requires_data_access, requires_systems_access, role_description_extended, role_name_former, role_name_local, role_subtype, succession_criticality, supports_initiatives, taxonomy_lineage, vacancy_duration

4.1.2 `integration` (378 entities)

Field	Populated	Sample Value
annual_cost	100.0%	{'amount': None, 'currency': 'USD', 'cost_components':
availability_sla	100.0%	{'target_uptime_pct': None, 'actual_uptime_pct': None,
crosses_boundary	100.0%	{'crosses_network_boundary': False, 'boundary_type': ''
data_exchanged	100.0%	{'data_description': '', 'data_classification': '', 'da
data_quality_score	100.0%	{'completeness_pct': None, 'accuracy_confidence': '', '
description	100.0%	Customer portal session authentication and token manage
direction	100.0%	Bidirectional
error_handling	100.0%	{'retry_mechanism': '', 'dead_letter_queue': False, 'al
frequency	100.0%	Real-Time

Field	Populated	Sample Value
integration_id	100.0%	INT-0001
integration_pattern	100.0%	Request-Response
integration_type	100.0%	API
last_major_change	100.0%	{'change_date': None, 'change_description': '', 'change
latency_requirement	100.0%	{'max_acceptable_ms': None, 'actual_p95_ms': None, 'mee
middleware_platform	100.0%	{'platform_name': '', 'platform_system_id': '', 'manage
name	100.0%	Identity Auth: Customer Portal (Fabric)
operational_status	100.0%	Active
provenance	100.0%	{'data_quality_score': {'completeness_pct': 32.0, 'accu
security_profile	100.0%	{'authentication': '', 'encryption_in_transit': False,
temporal	100.0%	{'effective_date': None, 'expiration_date': None, 'last

Unpopulated fields (0%): change_history, confidence_level, effective_date, integration_support_team, metadata, monitoring_status, owner, retirement_date, source_systems, target_systems, technical_debt_indicators, vendor

4.1.3 `control` (340 entities)

Field	Populated	Sample Value
applicability_dimensions	100.0%	{'people': False, 'process': False, 'technology': False
automation_details	100.0%	{'tool_name': '', 'system_id': '', 'configuration_refer
control_domain	100.0%	Access Control
control_effectiveness	100.0%	{'rating': '', 'methodology': '', 'last_assessed': None
control_id	100.0%	CTL-001
control_owner	100.0%	Identity Team
control_status	100.0%	Implemented
control_type	100.0%	Preventive
description	100.0%	MFA enforcement for all privileged, remote, and adminis
evidence_requirements	100.0%	{'evidence_types': [], 'evidence_location': '', 'retent
gap_status	100.0%	{'gap_exists': False, 'gap_description': '', 'remediati
kpi	100.0%	{'metric_name': '', 'target': '', 'current_value': '',
name	100.0%	Multi-Factor Authentication
provenance	100.0%	{'data_quality_score': {'completeness_pct': 35.0, 'accu
temporal	100.0%	{'effective_date': None, 'expiration_date': None, 'last

Field	Populated	Sample Value
testing_approach	100.0%	{'test_type': '', 'test_frequency': '', 'last_test_date

Unpopulated fields (0%): assessment_cadence, assessment_history, assessment_question, audit_findings, compensating_controls, control_category, control_class, control_operator, control_weighting, dependencies, exceptions, implementing_org_unit, implements_policies, maps_to_frameworks, maps_to_regulations, metadata, mitigates_risks, next_assessment_date, privacy_principle_mappings

4.2 Organizational Structure

Departments, organizational units, and persons model the human topology of the enterprise. The schema captures reporting lines, headcount, budget ownership, and geographic distribution at a level of specificity that supports analysis of decision authority concentration, single points of organizational failure, and talent dependency risk.

4.2.1 `department` (250 entities)

Field	Populated	Sample Value
budget	100.0%	5980000.0
code	100.0%	FIN
description	100.0%	Corporate finance function including financial planning
headcount	100.0%	46
name	100.0%	Finance Department
provenance	100.0%	{'data_quality_score': {'completeness_pct': 55.0, 'accu

Field	Populated	Sample Value
temporal	100.0%	{'effective_date': None, 'expiration_date': None, 'last

Unpopulated fields (0%): head_id, location_id, metadata, parent_department_id

4.2.2 `organizational_unit` (109 entities)

Field	Populated	Sample Value
acquisition_source	100.0%	{'source_entity_name': '', 'acquisition_date': None, 'd
attrition_rate	100.0%	{'annual_total_pct': None, 'voluntary_pct': None, 'invo
budget_authority	100.0%	{'approval_limit': None, 'currency': 'USD', 'delegation
center_of_excellence_details	100.0%	{'expertise_domains': [], 'standards_published': [], 'a
charter_document	100.0%	{'document_reference': '', 'document_date': None, 'last
control_environment_maturity	100.0%	{'overall_score': None, 'dimensions': [], 'assessed_dat
cost_structure	100.0%	{'annual_opex': None, 'annual_capex': None, 'total_annu
delegation_of_authority	100.0%	{'financial_limit': None, 'hiring_authority': False, 'c
description	100.0%	SEC-reportable segment. Service-centric, capital-light
employee_count	100.0%	{'fte': None, 'contractor': None, 'vendor_fte': None, '

Field	Populated	Sample Value
functional_domain_primary	100.0%	Service Delivery
governing_board	100.0%	{'board_name': '', 'board_type': '', 'member_count': No
integration_management_office_details	100.0%	{'integration_target': '', 'start_date': None, 'target_
joint_venture_details	100.0%	{'partner_entities': [], 'ownership_split': {'enterpris
legal_entity_details	100.0%	{'jurisdiction': '', 'registration_id': '', 'entity_typ
litigation_exposure	100.0%	{'active_matters_count': None, 'severity_assessment': '
market_position	100.0%	{'position_description': '', 'competitive_rank': None,
name	100.0%	Public Cloud Business Unit
operating_hours	100.0%	{'timezone_primary': '', 'timezones_all': [], 'operatin
operating_model	100.0%	Hybrid
organizational_health_score	100.0%	{'score': None, 'methodology': '', 'dimensions': [], 'a
parent_reporting_relationship	100.0%	{'reports_to_unit_id': '', 'reports_to_leader': '', 're
pl_accountability	100.0%	False
provenance	100.0%	{'data_quality_score': {'completeness_pct': 48.0, 'accu

Field	Populated	Sample Value
revenue_attribution	100.0%	{'annual_revenue': None, 'currency': 'USD', 'fiscal_yea
shared_service_details	100.0%	{'service_catalog': [], 'chargeback_model': '', 'client
span_of_control	100.0%	{'average_direct_reports': None, 'min_direct_reports':
temporal	100.0%	{'effective_date': None, 'expiration_date': None, 'last
unit_id	100.0%	OU-001
unit_type	100.0%	Business Unit

Unpopulated fields (0%): audit_findings, audit_scope, bu_fy2025_bookings_growth_target_pct, bu_fy2025_goals, bu_fy2025_kpis, bu_fy2025_revenue_growth_target_pct, bu_fy2025_revenue_target_usd, business_continuity_tier, business_criticality, compliance_officer, cost_breakdown, cost_center_id, criticality_justification, culture_integration_status, data_protection_officer, divestiture_candidacy, employee_count_by_location, financial_consolidation_entity, formation_date, functional_domain_secondary, geographic_presence, geographic_scope, governance_cadence, hierarchy_memberships, intercompany_relationships, key_person_dependencies, language_primary, languages_supported, leadership_team, management_layers, matrix_relationships, metadata, operational_status, operational_status_effective_date, operational_status_rationale, origin, osint_sources, profit_center_id, regulatory_environment, risk_exposure_inherent, risk_exposure_residual, risk_factors, sanctions_screening_status, statutory_reporting_obligations, strategic_alignment, strategic_role, strategic_role_rationale, tax_jurisdictions, taxonomy_lineage, transformation_stage, unit_description_extended, unit_leader, unit_leader_title, unit_name_former, unit_name_local, unit_subtype

4.2.3 `person` (100 entities)

Field	Populated	Sample Value
background_check_status	100.0%	{'status': '', 'check_type': [], 'completion_date': Non
confidence_score	13.0%	0.88
department	13.0%	Revenue Operations
description	28.0%	Former Managing Director, Asia Pacific and Japan at Rac
email	31.0%	ir@rackspace.com
employee_id	98.0%	EMP-001
employment_status	1.0%	Active
experience_profile	100.0%	{'total_years_professional': None, 'years_in_current_ro
first_name	100.0%	Gajen
hire_date	25.0%	2025-09-03
insider_status	100.0%	{'is_insider': False, 'insider_type': '', 'effective_da
is_active	100.0%	True
is_privileged	1.0%	True
job_title	1.0%	Chief Commercial Officer
last_name	100.0%	Kandiah
location	13.0%	San Antonio, TX
name	100.0%	Gajen Kandiah

Field	Populated	Sample Value
person_name	100.0%	{'display_name': '', 'given_name': '', 'family_name': '
provenance	100.0%	{'data_quality_score': {'completeness_pct': 50.0, 'accu
reporting_to	100.0%	{'manager_person_id': '', 'manager_role_id': '', 'repor
seniority_level	14.0%	C-Suite
source_attribution	13.0%	dept-175 Revenue Operations description, LinkedIn, Rack
temporal	100.0%	{'effective_date': None, 'expiration_date': None, 'last
title	99.0%	Chief Executive Officer

Unpopulated fields (0%): access_privileges, acquisition_origin, career_aspirations, certifications_held, clearance_level, clearances_held, conflict_of_interest_declarations, cost_center, current_roles, department_id, development_plan, dotted_line_to, education, employment_type, flight_risk, has_system_access, holds_roles, languages, located_at, location_primary, location_secondary, manages_vendor_relationships, mandatory_training_compliance, member_of_unit, mentored_by, mentors, metadata, organizational_unit_primary, organizational_unit_secondary, original_hire_date, participates_in_initiatives, performance_rating_current, performance_rating_history, performance_trajectory, person_id, phone, potential_assessment, regulatory_fitness, retention_actions, skills_gap_assessment, skills_inventory, subject_to_compliance, succession_candidate_for, timezone, training_completed, work_arrangement

4.3 Data and Information

Data assets and data domains are the most attribute-dense types in the graph. They exist because the CDAIO assessment premise is that enterprise data is an asset class, and asset classes require inventories with standardized attributes. The containment hierarchy (domains contain assets) models the logical structure a data governance program needs to function.

4.3.1 `data_asset` (164 entities)

Field	Populated	Sample Value
access_control_model	100.0%	RBAC
access_governance	100.0%	{'access_request_process': 'ServiceNow access request w
acquisition_cost	100.0%	{'cost': None, 'currency': 'USD', 'renewal_date': None}
acquisition_source	100.0%	{'source_name': '', 'acquisition_date': None, 'license_
ai_training_usage	100.0%	{'used_for_ai_training': False, 'model_references': [],
anonymization_status	100.0%	{'anonymized': False, 'anonymization_method': '', 're_i
archival_strategy	100.0%	{'hot_tier_retention': '', 'warm_tier_retention': '', '
asset_subtype	100.0%	Security Event & Log Aggregation Store
asset_type	100.0%	SIEM Index Cluster
backup_status	100.0%	{'backup_frequency': 'Daily', 'backup_type': 'Full + In
breach_notification_obligation	100.0%	{'applicable': True, 'notification_timeframe_hours': 72

Field	Populated	Sample Value
catalog_status	100.0%	{'cataloged': True, 'catalog_platform': 'ServiceNow CMD
change_reason	9.1%	KG enrichment — Phase 1 data asset completion
classification	100.0%	confidential
consent_management	100.0%	{'requires_consent': False, 'consent_type': '', 'consen
cross_border_transfer_status	100.0%	{'data_crosses_borders': True, 'source_jurisdictions':
current_access_grants	100.0%	{'total_users_with_access': None, 'roles_with_access':
data_classification	100.0%	confidential
data_custodian	85.4%	Splunk Platform Administrator
data_domain_primary	100.0%	Security Operations
data_format	89.0%	Semi-Structured
data_masking	100.0%	{'masking_applied': True, 'masking_type': 'Dynamic Mask
data_owner	85.4%	CISO / VP of Security Operations
data_steward	85.4%	Director of Security Engineering
data_subject_rights_applicable	100.0%	True
data_type	100.0%	semi-structured
description	100.0%	Primary security information and event management data

Field	Populated	Sample Value
dsar_process_reference	4.3%	Privacy team DSAR process — billing data subject to cus
effective_date	9.1%	2024-01-01
encryption_at_rest	100.0%	{'encrypted': True, 'algorithm': 'AES-256', 'key_manage
format	100.0%	Splunk Index
functional_domain_primary	100.0%	Security Operations
functional_domain_secondary	61.6%	['IT Operations', 'Compliance', 'DevOps']
golden_record_status	100.0%	{'is_golden_source': False, 'golden_source_reference':
is_encrypted	100.0%	True
last_review_date	9.1%	2025-10-01
lineage_completeness	85.4%	Substantially Documented
lineage_downstream	78.0%	[{'asset_id': '', 'asset_name': 'Microsoft Sentinel SIE
lineage_upstream	81.1%	[{'asset_id': '', 'asset_name': 'All Infrastructure & A
name	100.0%	Splunk Enterprise Security Index Cluster
next_review_date	9.1%	2026-10-01
origin	89.0%	Organic
partitioning_strategy	100.0%	{'partition_key': '', 'partition_type': ''}

Field	Populated	Sample Value
pii_categories	70.7%	['IP Addresses', 'Usernames', 'Authentication Tokens',
privacy_impact_assessment	100.0%	{'pia_required': True, 'pia_completed': True, 'pia_date
processing_cost	100.0%	{'annual_cost': None, 'currency': 'USD', 'cost_driver':
profiling_status	100.0%	{'last_profiled_date': None, 'profiling_tool': '', 'pro
provenance	100.0%	{'data_quality_score': {'completeness_pct': 62.0, 'accu
quality_monitoring	100.0%	{'automated_monitoring': False, 'monitoring_tool': '',
quality_score_composite	100.0%	{'score': None, 'scale': '1-5', 'assessed_date': None}
record_count_detail	100.0%	{'current_count': None, 'growth_rate_monthly_pct': None
regulations	24.4%	['SOC 2 Type II', 'PCI DSS', 'GDPR', 'FedRAMP']
regulatory_applicability	93.9%	[{'regulation_id': '', 'regulation_name': 'SOC 2 Type I
replication	100.0%	{'is_replicated': True, 'replication_type': 'Synchronou
retention_compliance	100.0%	{'policy_reference': 'Rackspace Log Retention Policy +
retention_days	84.8%	365
storage_cost	100.0%	{'annual_cost': None, 'currency': 'USD', 'cost_driver':

Field	Populated	Sample Value
storage_format	100.0%	{'format_type': 'Splunk TSIDX + Journal (Proprietary In
storage_technology	100.0%	{'system_id': '', 'system_name': 'Splunk Enterprise Sec
system_id	12.2%	sys-141
temporal	100.0%	{'effective_date': '2020-01-01', 'expiration_date': Non
third_party_sharing	0.6%	[{'third_party_name': 'Foundation Model Providers (Open
total_data_cost	100.0%	{'annual_total': None, 'currency': 'USD'}
value_assessment	100.0%	{'estimated_value': None, 'valuation_method': '', 'conf
velocity	100.0%	{'creation_rate': '', 'update_rate': '', 'deletion_rate
volume	100.0%	{'current_size': 110.0, 'size_unit': 'TB', 'growth_rate

Unpopulated fields (0%): asset_description_extended, asset_id, asset_name_common, attestation_status, audit_findings, cost_optimization_opportunities, data_cleansing_history, data_domain_secondary, fitness_for_purpose, indexing_strategy, metadata, owner_id, quality_dimensions, quality_issues_open, quality_rules, quality_trend, record_count

4.3.2 `data_domain` (88 entities)

Field	Populated	Sample Value
business_glossary_reference	100.0%	{'glossary_platform': '', 'glossary_url': '', 'term_cou

Field	Populated	Sample Value
data_classification	100.0%	Confidential
description	100.0%	Master data for 81,000+ customer accounts across 120 co
domain_id	100.0%	DD-001
domain_owner	100.0%	Chief Revenue Officer
domain_type	100.0%	Business
maturity_level	100.0%	Managed
monetization_potential	100.0%	{'potential_type': '', 'estimated_annual_value': None,
name	100.0%	Customer Account Data
provenance	100.0%	{'data_quality_score': {'completeness_pct': 45.0, 'accu
quality_targets	100.0%	{'completeness_target_pct': None, 'accuracy_target_pct'
retention_policy	100.0%	{'minimum_retention': 'Duration of customer relationshi
sensitivity_flags	100.0%	{'pii_flag': True, 'phi_flag': False, 'pci_flag': False
strategic_value	100.0%	Critical
temporal	100.0%	{'effective_date': None, 'expiration_date': None, 'last

Unpopulated fields (0%): data_residency_requirements, domain_scope, domain_steward, governing_policies, maturity_dimensions, metadata, regulatory_sensitivity, sub_domains

4.4 Systems and Technology

Systems, networks, sites, and locations model the physical and logical infrastructure that hosts, processes, and transmits enterprise data. These types support analysis of technology concentration risk, hosting dependency, and geographic distribution of processing capability.

4.4.1 `system` (164 entities)

Field	Populated	Sample Value
access_control_model	100.0%	RBAC
access_review_status	100.0%	{'last_review_date': None, 'next_review_date': None, 'r
acquisition_source	100.0%	{'source_entity_name': '', 'acquisition_date': None, 'i
api_surface	100.0%	{'api_count': None, 'api_types': ['REST'], 'api_documen
authentication_mechanisms	99.4%	[{'mechanism': 'SSO', 'protocol': 'SAML', 'mfa_supporte
availability	0.6%	high
availability_design	100.0%	{'architecture_pattern': 'Active-Active', 'redundancy_l
availability_sla	100.0%	{'target_uptime_pct': None, 'measurement_window': '', '
backup_status	100.0%	{'backup_frequency': 'Daily', 'backup_type': 'Full', 'l
business_fitness	100.0%	{'composite_score': None, 'dimensions': [], 'assessed_d
business_impact_if_unavailable	100.0%	{'impact_description': 'Complete loss of customer self-

Field	Populated	Sample Value
capacity_current	100.0%	{'cpu_utilization_pct': None, 'memory_utilization_pct':
capacity_headroom	100.0%	{'months_to_capacity': None, 'scaling_plan': '', 'scali
capex_remaining	100.0%	{'remaining_book_value': None, 'currency': 'USD', 'amor
change_velocity	100.0%	{'releases_per_month': None, 'deployment_frequency': ''
cloud_provider	0.6%	AWS
confidentiality	0.6%	high
containerized	100.0%	{'is_containerized': False, 'container_platform': '', '
contract_details	100.0%	{'vendor': '', 'contract_id': '', 'contract_start': Non
cost_per_transaction	100.0%	{'amount': None, 'currency': 'USD', 'basis': ''}
cost_per_user	100.0%	{'amount': None, 'currency': 'USD', 'basis': ''}
criticality	100.0%	critical
current_users	100.0%	{'total_licensed_users': None, 'total_provisioned_users
current_version_info	100.0%	{'version': '', 'release_date': None, 'patch_level': ''
cyber_exposure	100.0%	{'attack_surface_type': 'Internet Facing', 'threat_prof
deployment	2.4%	on-premises Kubernetes

Field	Populated	Sample Value
deployment_model	100.0%	on_premises
description	100.0%	Rackspace customer management portal built on Fabric au
encryption_profile	100.0%	{'data_at_rest': 'AES-256', 'data_in_transit': 'TLS 1.2
environment	100.0%	production
hostname	97.6%	portal.rackspace.com
incident_history	100.0%	{'p1_count_12m': None, 'p2_count_12m': None, 'mttr_hour
infrastructure_as_code	100.0%	{'iac_enabled': False, 'iac_tool': '', 'coverage_pct':
integration_points	0.6%	['prometheus', 'splunk', 'datadog', 'servicenow', 'slac
integrity	0.6%	critical
ip_address	97.6%	N/A
is_internet_facing	100.0%	True
license_details	100.0%	{'license_type': '', 'license_count': None, 'licenses_i
lifecycle_stage	97.6%	production
maintenance_windows	100.0%	{'schedule': '', 'duration_hours': None, 'impact_scope'
name	100.0%	Rackspace Customer Portal (Fabric)
observability_stack	100.0%	{'logging_tool': 'Splunk Enterprise Security', 'logging

Field	Populated	Sample Value
open_source_components	100.0%	{'component_count': None, 'license_types': [], 'sbom_av
os	97.6%	Linux
osint_confidence	2.4%	0.92
patch_compliance	100.0%	{'patch_policy': '', 'compliance_pct': None, 'days_to_p
penetration_test_status	100.0%	{'last_test_date': None, 'test_type': '', 'findings_cri
performance_sla	100.0%	{'response_time_target_ms': None, 'throughput_target':
provenance	100.0%	{'data_quality_score': {'completeness_pct': 40.0, 'accu
replacement_candidate	100.0%	{'has_replacement_planned': False, 'replacement_system_
scalability_profile	100.0%	{'horizontal_scaling': True, 'vertical_scaling': True,
security_architecture_review	100.0%	{'last_review_date': None, 'review_outcome': '', 'open_
support_model	100.0%	{'support_provider': 'Internal', 'support_tier': 'Tier
system_cost_benchmark	100.0%	{'benchmark_source': '', 'benchmark_value': None, 'perc
system_id	2.4%	sys-vault-001
system_type	97.6%	application
technical_fitness	100.0%	{'composite_score': None, 'dimensions': [], 'assessed_d

Field	Populated	Sample Value
temporal	100.0%	{'effective_date': None, 'expiration_date': None, 'last
tier	2.4%	T2
total_cost_of_ownership	100.0%	{'annual_tco': None, 'currency': 'USD', 'fiscal_year':
vendor	2.4%	HashiCorp (IBM)
vulnerability_profile	100.0%	{'last_scan_date': None, 'critical_count': None, 'high_

Unpopulated fields (0%): architecture_type, audit_findings, business_criticality, cloud_native_score, compliance_certifications, cost_breakdown, cost_optimization_opportunities, cost_trend, criticality_justification, data_classification_handled, data_residency_constraints, data_steward, data_stores, department_id, development_team, functional_domain_primary, functional_domain_secondary, governance_body, hosting_model, innovation_potential, metadata, monitoring_coverage, network_id, network_zone, operational_status, operational_status_rationale, origin, owner_id, ports, procurement_type, programming_languages, regulatory_applicability, risk_exposure_inherent, risk_exposure_residual, security_owner, strategic_alignment, strategic_classification, strategic_classification_rationale, support_team, system_category, system_description_extended, system_name_common, system_name_former, system_owner, system_subcategory, system_tier, taxonomy_lineage, technical_debt_indicators, technical_owner, technologies, technology_stack, user_base_by_geography, vendor_account_manager, vendor_id, version_currency

4.4.2 `network` (94 entities)

Field	Populated	Sample Value
cidr	100.0%	Various (4 IPv4 prefixes, 11,776 IPs originated)

Field	Populated	Sample Value
description	100.0%	[CONFIDENCE: T1] RADB as-name RACKSPACE. Global backbon
is_monitored	100.0%	True
name	100.0%	AS12200 Rackspace Global Backbone
provenance	100.0%	{'data_quality_score': {'completeness_pct': 45, 'accura
temporal	100.0%	{'effective_date': None, 'expiration_date': None, 'last
zone	100.0%	backbone

Unpopulated fields (0%): dns_servers, gateway, location_id, metadata, vlan_id

4.4.3 `site` (58 entities)

Field	Populated	Sample Value
address	100.0%	{'street_line_1': '1 Fanatical Place', 'street_line_2':
annual_operating_cost	100.0%	{'amount': None, 'currency': 'USD', 'fiscal_year': ''}
coordinates	100.0%	{'latitude': None, 'longitude': None}
current_occupancy	100.0%	{'headcount': None, 'as_of_date': None}
description	100.0%	Corporate HQ. Executive leadership, corporate functions
name	100.0%	Global Headquarters - San Antonio

Field	Populated	Sample Value
ownership_type	100.0%	Leased
provenance	100.0%	{'data_quality_score': {'completeness_pct': 60.0, 'accu
site_code	13.8%	SAT-HQ
site_id	100.0%	SITE-001
site_status	100.0%	Active
site_type	100.0%	Headquarters
strategic_designation	100.0%	Strategic
temporal	100.0%	{'effective_date': None, 'expiration_date': None, 'last

Unpopulated fields (0%): accessibility_compliance, acquisition_source, available_capacity, backup_site, book_value, building_count, business_continuity_tier, capacity_forecast, capital_investment_planned, climate_risk_assessment, connected_to_sites, cooling_capacity, cost_benchmark, cost_breakdown, cost_per_unit, crime_environment, deferred_maintenance_backlog, design_capacity, disaster_recovery_role, employs_persons, environmental_certifications, environmental_controls, evacuation_plan, expansion_potential, facility_condition_index, fire_suppression, floors, geopolitical_risk, governed_by, hosts_systems, houses_org_units, insurance_coverage, last_incident, lease_details, located_in_geography, managed_by_vendors, metadata, natural_hazard_profile, network_connectivity, occupancy_by_unit, origin, parking_capacity, physical_security_systems, physical_security_tier, power_supply, proximity_risks, replacement_value, risk_exposure_inherent, risk_exposure_residual, seasonal_variation, serves_customers_from, shift_model, single_points_of_failure, site_description, site_name_former, site_name_local, site_status_rationale, site_subtype, special_infrastructure, stores_data, subject_to_initiatives, subject_to_jurisdictions, total_area, usable_area, utilization_rate, waste_management, water_supply, year_built, year_last_renovated

4.4.4 `location` (46 entities)

Field	Populated	Sample Value
address	54.3%	19122 US Highway 281 N, Suite 128
city	100.0%	San Antonio
country	100.0%	US
description	100.0%	Corporate office at RidgeWood Plaza II, ~80,000 sq ft,
has_physical_security	100.0%	True
is_primary	100.0%	True
location_type	100.0%	office
name	100.0%	Global Headquarters - RidgeWood Plaza II
provenance	100.0%	{'confidence_level': 'High', 'assessed_by': 'AI Enrichm
security_level	100.0%	standard
state	80.4%	TX
temporal	100.0%	{'effective_date': None, 'expiration_date': None, 'last

Unpopulated fields (0%): capacity, metadata, zip_code

4.5 Market and External

No enterprise operates in isolation. Customers, vendors, regulations, jurisdictions, geographies, contracts, and market segments model the external forces that constrain and enable the enterprise. Regulatory and contractual constraints on data movement are as material to a CDAIO assessment as internal architecture.

4.5.1 `customer` (85 entities)

Field	Populated	Sample Value
account_tier	100.0%	Strategic
customer_id	100.0%	CUST-001
customer_type	100.0%	Enterprise
description	100.0%	Leading pediatric hospital in Seattle, Washington. Oper
name	100.0%	Seattle Children's Hospital
provenance	100.0%	{'data_quality_score': {'completeness_pct': None, 'accu
relationship_status	100.0%	Active
temporal	100.0%	{'effective_date': None, 'expiration_date': None, 'last

Unpopulated fields (0%): account_roles, account_team, audit_findings, belongs_to_segment, buys_products, communication_preferences, contract_value, credit_profile, customer_compliance_certifications, customer_locations, customer_name_common, customer_name_former, customer_risk_profile, customer_size, customer_subtype, customizations, data_privacy_obligations, data_shared_with, deployment_model, engagement_model, expansion_potential, export_control_status, functional_domain_primary, headquarters_location, health_score, industry_classification, integration_points, kyc_status, located_in, managed_by_org_unit, managed_through_vendors, metadata, origin, parent_customer, payment_terms, primary_contact, products_active, products_historical, profitability, regulatory_applicability, relationship_start_date, relationship_tenure_years, revenue_by_product, revenue_concentration_risk, revenue_current, revenue_lifetime, sanctions_screening, satisfaction_current, service_agreements, subsidiaries, uses_systems, wallet_share

4.5.2 `vendor` (55 entities)

Field	Populated	Sample Value
description	100.0%	Hyperscale public cloud platform provider; Rackspace Pu
has_data_access	100.0%	True
name	100.0%	Amazon Web Services, Inc.
primary_contact	78.2%	https://aws.amazon.com/contact-us/
provenance	100.0%	{'data_quality_score': {'completeness_pct': None, 'accu
risk_tier	100.0%	critical
temporal	100.0%	{'effective_date': None, 'expiration_date': None, 'last
vendor_id	100.0%	VND-001
vendor_type	100.0%	Hyperscale Cloud Platform

Unpopulated fields (0%): audit_findings, business_continuity, co_investment, compliance_certifications, contract_expiry, contract_value, cost_benchmark, cost_trend, cybersecurity_assessment, data_classification_access, data_processing, delivery_performance, dependency_on_vendor, diversity_classification, engagement_model, escalation_history, exclusivity, executive_sponsor, financial_stability, force_majeure_exposure, geographic_concentration, governance_cadence, headquarters_location, holds_contracts, incident_history, industry_classification, innovation_contribution, insurance_coverage, joint_governance, managed_by_org_unit, metadata, operates_from_locations, origin, ownership_structure, parent_company, partnership_structure, payment_terms, performance_dimensions, performance_scorecard, provides_data, provides_systems, quality_metrics, regulatory_applicability, relationship_start_date, relationship_status, relationship_tenure_years, sanctions_screening, savings_realized, serves_customers_through, shared_ip, sla_compliance, sla_uptime, spend_by_business_unit, spend_by_category, spend_concentration_risk, strategic_importance, subsidiaries, substitutability, supplies_products, sup-

ply_chain_role, total_annual_spend, vendor_category, vendor_compliance_certifications, vendor_dependencies, vendor_locations, vendor_management_roles, vendor_manager, vendor_name_common, vendor_name_former, vendor_risk_profile, vendor_size, vendor_subcategory

4.5.3 `regulation` (70 entities)

Field	Populated	Sample Value
applicability_criteria	100.0%	{'criteria_description': '', 'triggers': []}
applicability_status	100.0%	Applicable
compliance_status	100.0%	{'status': '', 'last_assessed': None, 'assessed_by': ''
description	100.0%	EU-wide data protection and privacy regulation governin
effective_date	100.0%	2018-05-25
issuing_body	100.0%	{'body_name': '', 'body_type': '', 'jurisdiction': ''}
monitoring_approach	100.0%	{'method': '', 'frequency': '', 'automated': False, 're
name	100.0%	General Data Protection Regulation
penalties	100.0%	{'penalty_type': '', 'maximum_penalty': '', 'penalty_ba
provenance	100.0%	{'data_quality_score': {'completeness_pct': 55.0, 'accu
regulation_category	100.0%	Data Privacy

Field	Populated	Sample Value
regulation_id	100.0%	REG-001
regulation_type	100.0%	Statutory
short_name	100.0%	GDPR
temporal	100.0%	{'effective_date': None, 'expiration_date': None, 'last

Unpopulated fields (0%): compliance_gaps, jurisdictions, key_requirements, last_amended_date, metadata, regulatory_change_pipeline

4.5.4 `jurisdiction` (61 entities)

Field	Populated	Sample Value
description	100.0%	US federal regulatory jurisdiction. SEC reporting (Exch
jurisdiction_code	100.0%	US
jurisdiction_id	100.0%	JUR-001
jurisdiction_type	100.0%	Federal
name	100.0%	United States Federal
provenance	100.0%	{'confidence_level': 'High', 'assessed_by': 'Thomas Jon
temporal	100.0%	{'effective_date': None, 'expiration_date': None, 'last

Unpopulated fields (0%): applicable_to_entities, child_jurisdictions, cross_border_transfer_rules, data_residency_requirements, governing_body, governs_capabilities, jurisdiction_name_local, labor_law_summary, links_to_compliance, metadata, parent_jurisdiction, privacy_regulation, regulatory_framework_summary, sanctions_status, sites_in_jurisdiction, tax_regime

4.5.5 `geography` (57 entities)

Field	Populated	Sample Value
description	100.0%	Primary operating region covering United States, Canada
geography_id	100.0%	GEO-001
geography_type	100.0%	Region
name	100.0%	North America
operational_status	100.0%	Active
provenance	100.0%	{'confidence_level': 'High', 'assessed_by': 'Thomas Jon
temporal	100.0%	{'effective_date': None, 'expiration_date': None, 'last

Unpopulated fields (0%): child_geographies, countries_included, geography_description, geography_name_short, market_characteristics, metadata, overlaps_with_jurisdictions, parent_geography, primary_languages, regional_leadership, sites_in_geography, strategic_importance, time_zones

4.5.6 `contract` (37 entities)

Field	Populated	Sample Value
contract_id	100.0%	230302
contract_status	100.0%	Active
contract_type	100.0%	Cooperative Purchasing Agreement
currency	100.0%	USD
description	100.0%	Cooperative purchasing agreement managed by The Interlo
end_date	86.5%	2026-05-31
name	100.0%	TIPS 230302 — Data Center Hosting, Sales and Service
provenance	100.0%	{'data_quality_score': {'completeness_pct': None, 'accu
start_date	100.0%	2023-05-31
temporal	100.0%	{'effective_date': None, 'expiration_date': None, 'last
total_contract_value	100.0%	0.0

Unpopulated fields (0%): amendments, annual_value, associated_contracts, auto_renewal, contract_owner, contracting_org_unit, covers_data, covers_products, covers_systems, current_term, data_handling_provisions, early_termination_penalty, governing_law, initial_term, insurance_requirements, ip_provisions, legal_owner, liability_caps, metadata, notice_period_days, opt_out_deadline, payment_schedule, payment_terms, procurement_owner, renewal_cap_pct, renewal_term, sla_summary, termination_for_convenience, termination_notice_days, termination_provisions, vendor_id, with_vendor

4.5.7 `market_segment` (26 entities)

Field	Populated	Sample Value
competitive_intensity	100.0%	High
description	100.0%	Large enterprise customers with 1000+ employees. Racksp
name	100.0%	Enterprise
provenance	100.0%	{'data_quality_score': {'completeness_pct': None, 'accu
segment_id	100.0%	SEG-001
segment_type	100.0%	Company Size
strategic_priority	100.0%	Primary
temporal	100.0%	{'effective_date': None, 'expiration_date': None, 'last

Unpopulated fields (0%): child_segments, contains_customers, entry_barriers, growth_rate, managed_by_org_unit, managing_org_unit, market_sizing, metadata, parent_segment, regulatory_environment, segment_criteria, segment_financial_summary, segment_level, segment_owner, served_by_products, win_rate

4.6 Strategic and Risk

Initiatives, risks, business capabilities, and incidents represent assessed conclusions rather than observed infrastructure. Their lower entity counts are intentional. Each connects to operational entities through typed relationships that preserve causal chains.

4.6.1 `initiative` (24 entities)

Field	Populated	Sample Value
annual_investment_usd	45.8%	150000000
annual_savings_target_usd	4.2%	100000000
current_status	100.0%	In Progress
description	100.0%	The foundational transformation of Rackspace's core bus
executive_sponsor	100.0%	Amar Maletira (CEO)
fiscal_year	100.0%	FY2025
functional_domain_primary	100.0%	Technology and Operations
initiative_category	100.0%	Strategic
initiative_dependencies	50.0%	['INIT-003', 'INIT-008', 'INIT-011']
initiative_id	100.0%	INIT-001
initiative_lead	100.0%	Dhrupad Trivedi (COO)
initiative_tier	100.0%	Portfolio
initiative_type	100.0%	Digital Transformation
name	100.0%	Multi-Cloud Managed Services Transformation
okrs_fy2025	100.0%	['Grow multi-cloud bookings to 70%+ of total new ARR',
phase	100.0%	Execution
planned_end_date	100.0%	2026-12-31
planned_start_date	100.0%	2021-01-01

Field	Populated	Sample Value
provenance	100.0%	{'data_quality_score': {'completeness_pct': None, 'accu
quarterly_milestones_fy2025	100.0%	{'Q1': 'APAC multi-cloud delivery model operational; Fa
rag_status	100.0%	Amber
strategic_priority	100.0%	Must Do
temporal	100.0%	{'effective_date': None, 'expiration_date': None, 'last

Unpopulated fields (0%): active_issues, active_risks, actual_end_date, actual_start_date, budget_breakdown, budget_variance, business_case_summary, child_initiatives, constraints, cost_benefit_analysis, critical_path_items, decision_log, dependencies_on_other_initiatives, expected_outcomes, forecast_at_completion, functional_domain_secondary, funding_source, governance_cadence, impacts_capabilities, impacts_customers, impacts_data, impacts_locations, impacts_org_units, impacts_products, impacts_roles, impacts_systems, impacts_vendors, initiative_description_extended, initiative_risk_profile, investment_thesis, key_milestones, lessons_learned, metadata, methodology, origin, owning_org_unit, parent_initiative, planned_duration_months, program_manager, related_initiatives, resource_allocation, resource_requirements, run_rate_impact, schedule_variance, spend_to_date, status_rationale, steering_committee, strategic_alignment, strategic_objectives, success_criteria, taxonomy_lineage, technology_requirements, total_budget, training_requirements, value_realized, vendor_engagement

4.6.2 `risk` (22 entities)

Field	Populated	Sample Value
acceptance_record	100.0%	{'accepted_by': '', 'acceptance_date': None, 'acceptanc

Field	Populated	Sample Value
assessment_cadence	100.0%	Quarterly
assessment_methodology	100.0%	Scenario Analysis
board_reportable	100.0%	True
control_effectiveness_on_risk	100.0%	{'risk_reduction_pct': None, 'control_count': None, 'we
description	100.0%	Risk of ransomware attack against Rackspace-managed inf
emerging_risk_flag	100.0%	False
inherent_financial_impact	100.0%	{'estimated_loss_low': None, 'estimated_loss_high': Non
inherent_impact	100.0%	Catastrophic
inherent_impact_dimensions	100.0%	{'financial': '', 'operational': '', 'reputational': ''
inherent_likelihood	100.0%	Likely
inherent_risk_level	100.0%	Critical
materiality_assessment	100.0%	{'pct_of_pretax_income': None, 'pct_of_total_assets': N
name	100.0%	Ransomware Attack on Hosted Exchange Infrastructure
nist_csf_function	100.0%	Respond
provenance	100.0%	{'data_quality_score': {'completeness_pct': 55.0, 'accu
residual_impact	100.0%	Major
residual_likelihood	100.0%	Possible
residual_risk_level	100.0%	High

Field	Populated	Sample Value
risk_appetite	100.0%	Averse
risk_category	100.0%	Cybersecurity
risk_id	100.0%	RSK-00001
risk_owner	100.0%	CISO
risk_source	100.0%	External
risk_status	100.0%	Mitigated
risk_subcategory	100.0%	Ransomware / Destructive Malware
risk_tolerance	100.0%	{'tolerance_threshold': '', 'escalation_trigger': '', '
risk_treatment	100.0%	Mitigate
risk_trend	100.0%	Stable
risk_velocity	100.0%	Instant
temporal	100.0%	{'effective_date': None, 'expiration_date': None, 'last
transfer_record	100.0%	{'transfer_mechanism': '', 'transfer_counterparty': '',
treatment_plan	100.0%	{'plan_description': '', 'target_residual_risk_level':

Unpopulated fields (0%): assessment_history, key_risk_indicators, last_assessed, loss_event_history, metadata, next_assessment, risk_group, risk_interconnections, risk_scenarios, risk_taxonomy_references

4.6.3 `business_capability` (64 entities)

Field	Populated	Sample Value
business_criticality	100.0%	Business Critical
capability_id	100.0%	BC-001
capability_type	100.0%	Business
description	100.0%	Enterprise-level strategic planning, long-term vision d
lifecycle_state	100.0%	Active
name	100.0%	Corporate Strategy & Planning
provenance	100.0%	{'data_quality_score': {'completeness_pct': 50.0, 'accu
temporal	100.0%	{'effective_date': None, 'expiration_date': None, 'last
tier	100.0%	Strategic

Unpopulated fields (0%): acquisition_source, audit_findings, business_impact_if_degraded, business_model_relevance, capability_description_extended, capability_name_local, capability_owner, capacity_utilization, child_capabilities, control_coverage, control_references, cost_benchmark, cost_to_operate, criticality_justification, cyber_exposure, delivered_through_products, depends_on_capabilities, depends_on_vendors, differentiation_evidence, differentiation_level, enables_capabilities, executive_sponsor, functional_domain, functional_domain_secondary, funding_amount, funding_model, governance_body, governance_cadence, governed_by, headcount_allocation, interfaces_with, kpi_current_values, kpi_definitions, lifecycle_next_state, lifecycle_next_state_target_date, lifecycle_state_rationale, lifecycle_transition_date, located_at, maturity_by_business_unit, maturity_by_region, maturity_composite_score, maturity_dimensions, maturity_model_reference, maturity_target, maturity_trajectory, metadata, origin, owned_by_organization, parent_capability, performance_status, raci_matrix, realized_by_ini-

tiatives, regulatory_applicability, resilience_tier, revenue_attribution, risk_exposure_inherent, risk_exposure_residual, risk_factors, rpo, rto, serves_customers, shared_service_model, single_points_of_failure, sourcing_suitability, staffed_by, strategic_alignment, strategic_risk_if_lost, supported_by_data, supported_by_systems, taxonomy_lineage, tier_justification, value_stream_alignment

4.6.4 `incident` (7 entities)

Field	Populated	Sample Value
description	100.0%	Complete outage of Rackspace Hosted Microsoft Exchange
detected_at	100.0%	2023-11-21
detection_method	100.0%	user_report
impact_description	100.0%	Complete outage of Rackspace Hosted Microsoft Exchange
incident_type	100.0%	malware
lessons_learned	100.0%	Patch velocity for critical Exchange vulnerabilities mu
name	100.0%	PLAY Ransomware Attack on Hosted Exchange (November 202
occurred_at	100.0%	2023-11-21
provenance	100.0%	{'confidence_level': 'High', 'assessed_by': 'AI Enrichm
resolved_at	100.0%	2024-03-31
root_cause	100.0%	PLAY ransomware group (Russia-linked threat actor) expl
severity	100.0%	critical

Field	Populated	Sample Value
status	100.0%	closed
temporal	100.0%	{'effective_date': None, 'expiration_date': None, 'last

Unpopulated fields (0%): affected_data_asset_ids, affected_system_ids, metadata, responder_ids, threat_actor_id

4.7 Products and Services

Product portfolios and products model the commercial output of the enterprise, connecting what the organization sells to the systems, data, and people that deliver it.

4.7.1 `product_portfolio` (71 entities)

Field	Populated	Sample Value
description	100.0%	SEC-reported revenue segment encompassing managed priva
lifecycle_stage	100.0%	Maturity
name	100.0%	Private Cloud
portfolio_id	100.0%	PF-001
portfolio_level	100.0%	L1
portfolio_type	100.0%	Service
provenance	100.0%	{'data_quality_score': {'completeness_pct': None, 'accu
strategic_role	100.0%	Core

Field	Populated	Sample Value
temporal	100.0%	{'effective_date': None, 'expiration_date': None, 'last

Unpopulated fields (0%): child_portfolios, contains_products, financial_summary, investment_priority, managed_by_org_unit, managing_org_unit, market_position, metadata, parent_portfolio, portfolio_owner, strategic_alignment

4.7.2 `product` (33 entities)

Field	Populated	Sample Value
description	100.0%	Proprietary multi-tenant software platform providing un
lifecycle_stage	100.0%	Maturity
name	100.0%	Rackspace Fabric
offering_subtype	100.0%	Platform
offering_type	100.0%	SaaS
product_id	100.0%	PRD-001
provenance	100.0%	{'data_quality_score': {'completeness_pct': None, 'accu
temporal	100.0%	{'effective_date': None, 'expiration_date': None, 'last

Unpopulated fields (0%): acquisition_source, belongs_to_portfolio, certifications_required, churn_rate, competitive_landscape, cost_to_deliver, customer_complaints, customer_count, customer_lifetime_value, customer_satisfaction, delivered_from_locations, delivery_channels, delivery_model, depends_on_vendors, dif-

ferentiation_factors, enabled_by_capabilities, enabled_by_systems, environmental_compliance, export_control_classification, fulfillment_model, functional_domain_primary, functional_domain_secondary, generation, launch_date, liability_exposure, lifecycle_stage_rationale, managed_by_org_unit, manufacturing_locations, margin_profile, market_position, metadata, origin, planned_retirement_date, pricing_model, pricing_range, product_category, product_description_internal, product_name_former, product_name_internal, product_name_local, product_reviews, product_roles, quality_metrics, recall_history, regulatory_applicability, regulatory_approval_status, regulatory_classification, related_products, revenue_by_geography, revenue_by_segment, revenue_current, revenue_trend, safety_certifications, scalability_constraints, serves_customers, service_level_performance, service_level_targets, sku_count, strategic_alignment, supply_chain_complexity, taxonomy_lineage, uses_data, version_current, version_history

4.8 Governance

Policies model the formal governance instruments that define expected behavior. The gap between policy count (114) and enforcement relationship count (386) indicates that enforcement mechanisms are modeled at a higher granularity than the policies they implement.

4.8.1 `policy` (114 entities)

Field	Populated	Sample Value
applies_to	100.0%	{'org_units': [], 'roles': [], 'systems': [], 'data_dom
communication_plan	100.0%	{'last_communicated': None, 'communication_channel': ''
compliance_measurement	100.0%	{'metric': '', 'target': '', 'current_value': '', 'meas
current_version	100.0%	{'version': '', 'effective_date': None, 'change_summary

Field	Populated	Sample Value
description	100.0%	Top-level ISMS policy establishing Rackspace Technology
enforcement_mechanism	100.0%	{'mechanism_type': '', 'description': '', 'automated':
framework	100.0%	ISO 27001
is_enforced	100.0%	True
name	100.0%	Information Security Management Policy
policy_id	100.0%	POL-001
policy_type	100.0%	Governance
provenance	100.0%	{'data_quality_score': {'completeness_pct': 40.0, 'accu
review_frequency_days	100.0%	365
severity	100.0%	critical
temporal	100.0%	{'effective_date': None, 'expiration_date': None, 'last
training_requirement	100.0%	{'required': False, 'training_program': '', 'completion

Unpopulated fields (0%): applicable_systems, approval_date, approving_authority, control_id, driven_by_regulations, driven_by_risks, effective_date, exceptions, last_reviewed_date, metadata, next_review_date, owner_id, policy_author, policy_owner, policy_requirements, policy_scope, policy_status, related_policies, review_cadence, version_history

5 Relationship Architecture

The graph encodes 7,466 relationships across 42 types, forming 114 distinct edge patterns (unique combinations of source entity type, relationship type, and target entity type). This chapter documents every relationship type, organized by functional category.

5.1 Base Relationship Schema

Every relationship carries a common set of attributes:

Attribute	Description	Population
`id`	Unique relationship identifier	100%
`relationship_type`	One of 42 defined types	100%
`source_id`	ID of the source entity	100%
`target_id`	ID of the target entity	100%
`weight`	Edge weight (0.0 to 1.0)	~100%
`confidence`	Confidence score (0.0 to 1.0)	~100%
`description`	Optional prose description	Varies
`properties`	Optional structured metadata	Varies

The `weight` field encodes the strength or significance of the relationship. The `confidence` field encodes the assessed reliability of the relationship's existence. Both default to 1.0 when not explicitly set, meaning absence of a confidence score does not indicate low confidence but rather that confidence has not been independently assessed.

5.2 Spatial and Jurisdictional

Geographic and jurisdictional placement are modeled as first-class relationships rather than entity attributes. This design decision enables queries about data residency compliance, regulatory exposure by jurisdiction, and operational concentration without requiring secondary attribute lookups. The high edge count (1,943 combined) reflects the enterprise's global operational footprint.

5.2.1 `located_in` (1260 edges)

Source Type	Target Type	Count
role	department	416
department	organizational_unit	250
data_asset	system	157
network	network	102
customer	geography	81
site	geography	58
geography	geography	53
location	geography	46
network	geography	37
contract	geography	26

Metadata coverage: description: 73.8%

5.2.2 `located_at` (683 edges)

Source Type	Target Type	Count
system	site	484
site	location	104
network	site	52
organizational_unit	site	41
data_asset	site	2

Metadata coverage: description: 0.3%, properties: 8.5%

5.3 Dependency and Constraint

These relationships form the analytical backbone of the graph. When a system depends_on another system that is hosted_on infrastructure subject_to a regulation, the graph can traverse the full obligation chain in a single query. This is where blast radius analysis and change impact assessment operate.

5.3.1 `subject_to` (1120 edges)

Source Type	Target Type	Count
data_domain	regulation	207
data_domain	policy	183
data_asset	data_domain	176
data_domain	jurisdiction	137
site	jurisdiction	80
regulation	jurisdiction	72
location	jurisdiction	68
customer	regulation	39
jurisdiction	jurisdiction	37
product	regulation	18

Metadata coverage: description: 74.6%

5.3.2 `depends_on` (790 edges)

Source Type	Target Type	Count
integration	system	463
system	system	213
integration	network	64

Source Type	Target Type	Count
business_capability	system	50

Metadata coverage: description: 10.3%

5.3.3 `hosted_on` (490 edges)

Source Type	Target Type	Count
system	site	489
system	system	1

Metadata coverage: properties: 99.8%

5.3.4 `applies_to` (543 edges)

Source Type	Target Type	Count
control	system	543

Metadata coverage: properties: 1.8%

5.4 Governance and Control

The distinction between enforces (386 edges) and governed_by (12 edges) is structurally significant. The graph contains extensive modeling of which controls enforce which policies, but sparse modeling of formal data governance accountability. This gap reflects the enterprise's actual governance maturity, not a modeling omission.

5.4.1 `enforces` (386 edges)

Source Type	Target Type	Count
control	policy	386

5.4.2 `managed_by` (385 edges)

Source Type	Target Type	Count
system	department	162
system	person	160
data_domain	department	63

Metadata coverage: description: 16.4%

5.4.3 `implements` (178 edges)

Source Type	Target Type	Count
policy	regulation	155
product	product_portfolio	9
product	system	6
product_portfolio	system	5
product_portfolio	product_portfolio	3

Metadata coverage: description: 12.9%

5.4.4 `governs` (32 edges)

Source Type	Target Type	Count
contract	customer	20
contract	system	5
contract	data_asset	4
contract	vendor	3

Metadata coverage: description: 100.0%

5.4.5 `governed_by` (12 edges)

Source Type	Target Type	Count
organizational_unit	policy	10
policy	regulation	2

5.5 Data Flow

These relationships model how data moves through the enterprise. Combined with integration entities (which carry their own rich attribute schemas), these edges create a queryable data lineage layer that supports both regulatory compliance analysis and operational dependency mapping.

5.5.1 `stores` (165 edges)

Source Type	Target Type	Count
system	data_asset	165

Metadata coverage: properties: 99.4%

5.5.2 `connects_to` (140 edges)

Source Type	Target Type	Count
integration	system	88
integration	data_asset	52

5.5.3 `integrates_with` (101 edges)

Source Type	Target Type	Count
system	system	101

Metadata coverage: properties: 99.0%

5.5.4 `processes` (55 edges)

Source Type	Target Type	Count
vendor	data_asset	55

Metadata coverage: description: 69.1%

5.6 Organizational

These connect people to their functions, departments to their parent units, and data domains to their constituent assets. The low count on reports_to (9) compared to has_role (92) reflects a deliberate design choice: the graph prioritizes functional authority mapping over hierarchical org-chart modeling.

5.6.1 `has_role` (92 edges)

Source Type	Target Type	Count
person	role	92

Metadata coverage: description: 83.7%, properties: 16.3%

5.6.2 `works_in` (48 edges)

Source Type	Target Type	Count
person	department	48

Metadata coverage: description: 43.8%, properties: 56.2%

5.6.3 `reports_to` (9 edges)

Source Type	Target Type	Count
person	person	9

Metadata coverage: properties: 100.0%

5.6.4 `staffed_by` (3 edges)

Source Type	Target Type	Count
organizational_unit	person	3

Metadata coverage: properties: 100.0%

5.6.5 `belongs_to` (71 edges)

Source Type	Target Type	Count
product	product_portfolio	71

Metadata coverage: properties: 100.0%

5.6.6 `holds` (37 edges)

Source Type	Target Type	Count
customer	contract	37

Metadata coverage: properties: 100.0%

5.6.7 `contains` (75 edges)

Source Type	Target Type	Count
data_domain	data_asset	51
market_segment	customer	21
product_portfolio	product	3

Metadata coverage: description: 96.0%

5.7 Commercial

Market-facing relationships connect the enterprise to its customers, vendors, and service delivery mechanisms. These support analysis of revenue concentration, vendor dependency, and contract-based constraints on operational flexibility.

5.7.1 `serves` (137 edges)

Source Type	Target Type	Count
system	customer	40
product	market_segment	39
product_portfolio	customer	32
organizational_unit	market_segment	26

Metadata coverage: description: 52.6%, properties: 41.6%

5.7.2 `provides_service` (71 edges)

Source Type	Target Type	Count
vendor	department	50
system	organizational_unit	21

Metadata coverage: properties: 100.0%

5.7.3 `buys` (22 edges)

Source Type	Target Type	Count
customer	product	22

5.7.4 `contracts_with` (7 edges)

Source Type	Target Type	Count
contract	vendor	7

Metadata coverage: properties: 100.0%

5.7.5 `supplies` (6 edges)

Source Type	Target Type	Count
vendor	product_portfolio	5
vendor	product	1

Metadata coverage: description: 100.0%

5.7.6 `supplied_by` (28 edges)

Source Type	Target Type	Count
system	vendor	28

Metadata coverage: properties: 100.0%

5.7.7 `delivers` (33 edges)

Source Type	Target Type	Count
system	product	33

Metadata coverage: properties: 100.0%

5.8 Risk and Impact

Risk relationships connect assessed risk entities to the operational entities they affect. The causal chain is preserved: a risk creates_risk for a system, which impacts a business capability, which affects a product portfolio. This enables multi-hop impact analysis.

5.8.1 `mitigates` (29 edges)

Source Type	Target Type	Count
control	risk	29

Metadata coverage: properties: 96.6%

5.8.2 `creates_risk` (4 edges)

Source Type	Target Type	Count
vendor	risk	4

Metadata coverage: properties: 100.0%

5.8.3 `impacts` (14 edges)

Source Type	Target Type	Count
initiative	business_capability	10
initiative	risk	4

Metadata coverage: properties: 85.7%

5.8.4 `impacted_by` (17 edges)

Source Type	Target Type	Count
business_capability	risk	6

Source Type	Target Type	Count
system	risk	6
system	incident	3
product	risk	2

Metadata coverage: properties: 100.0%

5.8.5 `affects` (21 edges)

Source Type	Target Type	Count
incident	system	19
incident	data_asset	2

5.8.6 `addresses` (15 edges)

Source Type	Target Type	Count
initiative	risk	15

Metadata coverage: properties: 73.3%

5.9 Strategic

Strategic relationships connect initiatives to the capabilities they enable, the budgets that fund them, and the systems they support. These provide the forward-looking dimension of the graph.

5.9.1 `supports` (209 edges)

Source Type	Target Type	Count
product_portfolio	business_capability	59
business_capability	department	52

Source Type	Target Type	Count
system	business_capability	36
business_capability	organizational_unit	24
vendor	product_portfolio	18
policy	policy	12
vendor	system	5
product_portfolio	organizational_unit	3

Metadata coverage: description: 77.0%, properties: 17.2%

5.9.2 `enables` (33 edges)

Source Type	Target Type	Count
product	business_capability	33

Metadata coverage: properties: 100.0%

5.9.3 `drives` (28 edges)

Source Type	Target Type	Count
initiative	business_capability	20
initiative	product	6
initiative	control	2

Metadata coverage: properties: 100.0%

5.9.4 `funded_by` (12 edges)

Source Type	Target Type	Count
initiative	organizational_unit	12

Metadata coverage: properties: 100.0%

5.9.5 `subject_of` (30 edges)

Source Type	Target Type	Count
customer	data_asset	30

Metadata coverage: description: 100.0%

5.9.6 `hosts` (26 edges)

Source Type	Target Type	Count
site	system	26

5.9.7 `regulates` (49 edges)

Source Type	Target Type	Count
jurisdiction	geography	49

6 Edge Pattern Analysis

The 114 distinct edge patterns represent the full vocabulary of structural relationships the graph can express.

6.1 Relationship Volume Distribution

The top 10 relationship types by edge count:

Relationship Type	Count	% of Total	Most Common Pattern
located_in	1260	16.9%	role -> department
subject_to	1120	15.0%	data_domain -> regulation
depends_on	790	10.6%	integration -> system
located_at	683	9.1%	system -> site
applies_to	543	7.3%	control -> system
hosted_on	490	6.6%	system -> site
enforces	386	5.2%	control -> policy
managed_by	385	5.2%	system -> department
supports	209	2.8%	product_portfolio -> business_capability
implements	178	2.4%	policy -> regulation

6.2 Entity Type Connectivity

Which entity types participate as sources and targets across the most relationship types, indicating structural centrality:

Entity Type	Outbound Rel Types	Inbound Rel Types	Total Edges
system	12	11	3624
product	6	5	243

Entity Type	Outbound Rel Types	Inbound Rel Types	Total Edges
data_asset	3	7	694
organizational_unit	4	4	390
customer	5	3	322
product_portfolio	4	4	211
vendor	5	3	176
policy	3	4	760
business_capability	3	4	290
department	1	5	1041
site	4	2	1336
person	3	3	321
risk	0	5	66
network	2	2	357
contract	3	1	102
data_domain	3	1	817
regulation	1	3	493
control	3	1	960
initiative	4	0	69
geography	1	2	403
location	2	1	218
jurisdiction	2	1	480
role	1	1	508
integration	2	0	667
market_segment	1	1	86

Entity Type	Outbound Rel Types	Inbound Rel Types	Total Edges
incident	1	1	24

7 Due Diligence Analysis Layer

The 401 structured JSON analysis files organized across nine due diligence stages represent the analytical layer built on top of the graph. The graph provides structural truth; the analysis layer provides assessed truth.

Stage	Name	Scope
Stage 0	Diligence Governance	Defines the constitutional rules and constraints governing the entire assessment methodology.
Stage 1	Corporate Legal Structural	Corporate legal existence, ownership and control structures, structural debt from historical M&A, change-of-control constraints, and jurisdictional licensing requirements.
Stage 2	Business Model Economics	Revenue engines, cost structure and margin physics, value flow friction, customer segmentation and dependency, partner and channel value capture, and business model coherence contradictions.
Stage 3	Market Competitive Force	Market structure and power migration, competitive threat vectors, customer buying and switching inertia, pricing power and discounting signals, partner and platform dependency, and external shock sensitivity.
Stage 4	Operating Model Execution	Decision rights and authority, operating cadence and throughput, cross-functional bottlenecks, incentive and accountability structures, and talent deployment constraints.
Stage 5	Financial	Revenue quality and stability, margin physics, liquidity and cash runway, debt covenants and control transfer triggers, capital allocation capacity, and financial coherence contradictions.

Stage	Name	Scope
	Capital Stress	
Stage 6	Technology Platform	Compute and datacenter control, cloud and hyperscaler dependency, platform criticality and fragility, vendor lock-in and licensing, technical debt constraints, and technology estate coherence.
Stage 7	Data Information Epistemic	Data domain authoritativeness, data lineage and meaning drift, data quality and error tolerance, stewardship accountability, access and privilege asymmetry, data flow latency, and epistemic coherence.
Stage 8	Risk Security Regulatory	Cybersecurity control failure surfaces, privacy and data use limits, regulatory enforcement exposure, data residency and sovereignty, and governance power and veto rights.
Stage 9	Resilience ESG Continuity	Business continuity and failure sequencing, workforce and talent fragility, vendor ecosystem shock propagation, insurance and risk transfer, and ESG materiality constraints.

7.1 Analysis File Structure

Each due diligence stage follows a consistent internal structure. Substages contain hypothesis files, disconfirming evidence searches, uncertainty registers, and domain-specific analysis artifacts. Each stage concludes with a validation folder containing contradiction analysis, exit criteria tests, gate decisions, and hypothesis discipline audits.

This structure encodes the analytical methodology itself, not just findings. The hypothesis-disconfirmation-validation cycle ensures that conclusions are stress-tested against contradicting evidence before they advance to the next stage.

8 Known Structural Gaps

Three structural gaps in the current graph constrain specific categories of analysis.

8.1 Data Quality Targets

The graph contains zero data quality target entities. It can describe the current state of data quality through composite scores on data asset entities, but it cannot model the gap between current state and target state. For a CDAIO assessment, that gap analysis is where actionable recommendations originate. Adding a `quality_target` entity type with relationships to `data_asset` and `data_domain` entities would close this gap.

8.2 Governance Relationship Density

The governance relationship layer is thin relative to the operational layer. Twelve `governed_by` edges across 252 data entities (88 domains plus 164 data assets) represents a 4.8% governance linkage rate. This limits the graph's ability to model accountability chains for data stewardship. The gap reflects the state of the enterprise being modeled rather than a schema limitation: the schema supports `governed_by` relationships, but the enterprise has not established them.

8.3 Temporal Window Coverage

The `valid_from` and `valid_until` base attributes are unpopulated across all 2,944 entities, meaning the graph represents a current-state snapshot rather than a time-series. Historical state analysis (such as tracking governance maturity improvement over quarters) would require populating these fields across future graph iterations.